

EAB 770 - Chapter 2

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1 Chapter 2: 2×2 tables

1.1 Outline

- Contingency Tables
- Chi-Square Statistics

- Exact Tests
- Difference in Proportions
- Odds Ratio and Relative Risk
- Sensitivity and Specificity
- McNemar's Test
- Incidence Densities
- Power Analysis

2 Contingency Tables

2.1 2x2 Contingency Tables

- Summarizes categorical data
 - Response variable has two levels
 - Exploratory variable has two levels
- Example:
 - Treatment/Placebo in RCT
 - Case/Control in a retrospective study

Risk\Case	Case	Control	Total
Present			
Absent			
Total			

2.2 Contingency Tables

We wish to determine whether there is an association between the row and column variables using the frequencies observed in each category.

i Note

Is the risk variable associated with the prevalence rate?

Recall: Intro to Statistics

- Chi-square tests of homogeneity H_0 : The two populations (rows) have the same distribution of the column variable
- Chi-square tests of independence H_0 : The row and column variables are independent of each other.

2.3 Contingency Tables: Convention

The indices of contingency tables denote the row and column of each number. For a total sample size of n ,

- n_{xy} denotes the number at row x and column y , also referred to as a joint frequency.
- Seeing a (+) sign indicates that it is a marginal sum, also referred to as marginal frequencies.
 - n_{x+} denotes the sum of all columns in row x .
 - n_{+y} denotes the sum of all rows in column y .
- The placement of the + sign denotes the margin in which it was summed.

Row\Col	C ₁	C ₂	Total
R ₁	n_{11}	n_{12}	n_{1+}
R ₂	n_{21}	n_{22}	n_{2+}
Total	n_{+1}	n_{+2}	n

2.4 SAS Example: Contingency Table

```
data respire;
length treat $7. outcome $1. ;
input treat $ outcome $ count;
datalines;
placebo f 16
placebo u 48
test f 40
test u 20
;
run;
```

```
proc freq data=respire order=data;
weight count;
tables treat*outcome;
run;
```

- `proc freq` calls the FREQ procedure in SAS.
 - Not defining `data=respire` means SAS will automatically use the most recently run data set.
- `order=data` means we are creating the rows and columns based on how the data was inputted, not alphabetical order.
- `weight` tells SAS the number of counts for each count
- `tables` creates the contingency table with row, column, and total percentages.
- `run;` tells SAS to run the code. **This is important!**

2.5 Contingency Tables in SAS- Long Format

When the data is not aggregated, the FREQ procedure can still be used to create contingency tables. We can use the import data procedure in SAS ODA to create `work.spore` from `Spore.csv` uploaded on Canvas.

```
proc freq order=data data=severe;
weight count;
tables treat*outcome / chisq nocol;
exact or;
run;
```

2.6 Contingency Tables in R

To create your own contingency table, you can use the `data.frame()` and `xtabs()` functions.

```
count_data <- data.frame(Row = c("A","A","B","B"),
                        Column = c("X","Y","X","Y"),
                        Count = c(2,4,6,8))
count_data
```

	Row	Column	Count
1	A	X	2
2	A	Y	4
3	B	X	6
4	B	Y	8

```
xtabs(Count~Row+Column,data=count_data)
```

```
      Column
Row X Y
A  2 4
B  6 8
```

You can also use the `expand.grid()` function to create combinations of variable levels.

```
count_data <- expand.grid(Row = factor(c("A","B"), levels=c("A","B")),
                        Column = factor(c("X","Y"), levels=c("X","Y")))

count_data$Count = c(2,4,6,8)

count_data
```

```
  Row Column Count
1   A      X     2
2   B      X     4
3   A      Y     6
4   B      Y     8
```

```
xtabs(Count~Row+Column,data=count_data)
```

```
      Column
Row X Y
A  2 6
B  4 8
```

Note

`ordered` can also be used to account for ordering in the row and column variables.

2.7 Contingency Tables in R- Long Format

When the data is not aggregated, the `xtabs` function can still be used to create contingency tables. First, we import `Spore.csv` from Canvas.

```
spore <- read.csv("Spore.csv",stringsAsFactors = TRUE)
spore
```

	ID	Treatment	Period	Spore
1	1	Test	Pre-Treatment	Full
2	1	Test	Mid-Treatment	Some
3	1	Test	Post-Treatment	Some
4	2	Test	Pre-Treatment	Full
5	2	Test	Mid-Treatment	None
6	2	Test	Post-Treatment	None
7	3	Test	Pre-Treatment	Full
8	3	Test	Mid-Treatment	Some
9	3	Test	Post-Treatment	Some
10	4	Test	Pre-Treatment	Full
11	4	Test	Mid-Treatment	Some
12	4	Test	Post-Treatment	
13	5	Test	Pre-Treatment	Full
14	5	Test	Mid-Treatment	None
15	5	Test	Post-Treatment	None
16	1	Placebo	Pre-Treatment	Full
17	1	Placebo	Mid-Treatment	Full
18	1	Placebo	Post-Treatment	Full
19	2	Placebo	Pre-Treatment	Full
20	2	Placebo	Mid-Treatment	Some
21	2	Placebo	Post-Treatment	Some
22	3	Placebo	Pre-Treatment	Full
23	3	Placebo	Mid-Treatment	Some
24	3	Placebo	Post-Treatment	Some
25	4	Placebo	Pre-Treatment	Full
26	4	Placebo	Mid-Treatment	Some
27	4	Placebo	Post-Treatment	Some
28	5	Placebo	Pre-Treatment	Some
29	5	Placebo	Mid-Treatment	
30	5	Placebo	Post-Treatment	
31	6	Test	Pre-Treatment	Full
32	6	Test	Mid-Treatment	Some
33	6	Test	Post-Treatment	Some
34	7	Test	Pre-Treatment	Full
35	7	Test	Mid-Treatment	None
36	7	Test	Post-Treatment	None
37	8	Test	Pre-Treatment	Full
38	8	Test	Mid-Treatment	Some

39	8	Test	Post-Treatment	Some
40	9	Test	Pre-Treatment	Full
41	9	Test	Mid-Treatment	Some
42	9	Test	Post-Treatment	None
43	10	Test	Pre-Treatment	Full
44	10	Test	Mid-Treatment	Some
45	10	Test	Post-Treatment	Some
46	6	Placebo	Pre-Treatment	Full
47	6	Placebo	Mid-Treatment	Full
48	6	Placebo	Post-Treatment	Full
49	7	Placebo	Pre-Treatment	Full
50	7	Placebo	Mid-Treatment	Full
51	7	Placebo	Post-Treatment	Full
52	8	Placebo	Pre-Treatment	Full
53	8	Placebo	Mid-Treatment	Full
54	8	Placebo	Post-Treatment	Full
55	9	Placebo	Pre-Treatment	Full
56	9	Placebo	Mid-Treatment	Some
57	9	Placebo	Post-Treatment	Some
58	10	Placebo	Pre-Treatment	Full
59	10	Placebo	Mid-Treatment	Some
60	10	Placebo	Post-Treatment	Some
61	11	Test	Pre-Treatment	Full
62	11	Test	Mid-Treatment	Some
63	11	Test	Post-Treatment	None
64	12	Test	Pre-Treatment	Full
65	12	Test	Mid-Treatment	Some
66	12	Test	Post-Treatment	None
67	13	Test	Pre-Treatment	Full
68	13	Test	Mid-Treatment	Some
69	13	Test	Post-Treatment	Some
70	14	Test	Pre-Treatment	Full
71	14	Test	Mid-Treatment	None
72	14	Test	Post-Treatment	None
73	15	Test	Pre-Treatment	Full
74	15	Test	Mid-Treatment	None
75	15	Test	Post-Treatment	None
76	11	Placebo	Pre-Treatment	Full
77	11	Placebo	Mid-Treatment	Some
78	11	Placebo	Post-Treatment	Some
79	12	Placebo	Pre-Treatment	Full
80	12	Placebo	Mid-Treatment	Some
81	12	Placebo	Post-Treatment	Some

```

82 13  Placebo  Pre-Treatment  Full
83 13  Placebo  Mid-Treatment  Some
84 13  Placebo  Post-Treatment Some
85 14  Placebo  Pre-Treatment  Full
86 14  Placebo  Mid-Treatment  Full
87 14  Placebo  Post-Treatment Full
88 15  Placebo  Pre-Treatment  Full
89 15  Placebo  Mid-Treatment  Full
90 15  Placebo  Post-Treatment Full

```

```
xtabs(~Treatment+Spore,data=spore)
```

	Spore			
Treatment	Full	None	Some	
Placebo	2	26	0	17
Test	1	15	13	16

2.8 Exercise

Suppose a survey asking sophomores and senior students whether they agree with the statement “I enjoyed my in-person classes this semester.” yielded the following results.

Enter the data in SAS using the data step and generate the contingency tables. Name the data set `enjoy`.

	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
Sophomore	24	43	12	25	12
Seniors	19	37	24	32	21

2.8.1 R Code

OR

```

enjoy <- expand.grid(year=factor(c("Sophomore","Seniors"), levels = c("Sophomore","Seniors")),
                    response = factor(c("SA","A","N","D","SD"),levels = c("SA","A","N","D","SD")),
enjoy$Count <- c(24,19,43,37,12,24,25,32,12,21)
xtabs(Count~year+response,data=enjoy)

```

	response				
year	SA	A	N	D	SD
Sophomore	24	43	12	25	12
Seniors	19	37	24	32	21

2.8.2 SAS Code

There are many ways to code the `data` step. Here's one way to do it!

```
data enjoy;
length year $9. response $2.;
input year $ response $ count @@;
datalines;
sophomore SA 24 sophomore A 43
sophomore N 12
sophomore D 25 sophomore SD 12
senior SA 19 senior A 37
senior N 24
senior D 32 senior SD 21
;

proc freq order=data data=enjoy;
weight count;
tables year*response;
run;
```

2.9 Independence

If the two categorical variables are independent of each other, they are not associated.

Note

Independence also means that the joint probability p_{ij} can be determined solely on the basis of the marginal probabilities. These can be estimated using sample data.

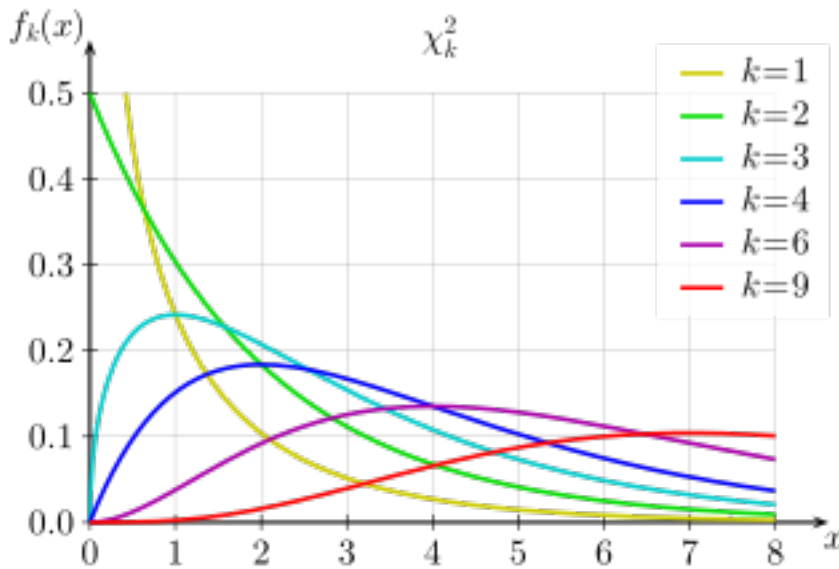
3 Chi-square Statistics

3.1 Chi-square Statistics

The null hypothesis of no association (independence or homogeneity) between row and column variables imply that the totals are fixed. The probabilities of realization of each cell is determined by the hypergeometric distribution.

3.2 Chi-square distribution

- The chi-square distribution, χ_k^2 , is defined by the number of degrees of freedom k , where $k \geq 1$.
- Properties include:
 - The value of χ^2 can only assume non-negative values.
 - The χ^2 distribution is asymmetric.
 - The sum of two or more independent chi-square variables also follows a chi-square distribution.
 - Mean: k , Variance: $2k$



3.3 Pearson Chi-square Statistic

The Pearson Chi-square statistic can be written in terms of the joint frequency n_{ij} and the expected frequency m_{ij} .

$$Q_P = \sum_{i=1}^2 \sum_{j=1}^2 \frac{(n_{ij} - m_{ij})^2}{m_{ij}}$$

i Note

m_{ij} is the expected value of the number of observations in row i and column j , n_{ij} . The expected frequency m_{ij} can be calculated based on the null hypothesis assumption.

- Independence: $p_{ij} = p_{i+}p_{+j} \rightarrow m_{ij} = \frac{n_{i+}n_{+j}}{n}$
- Homogeneity: $p_{i+} = n_{i+}/n; p_{+j} = n_{+j}/n \rightarrow m_{ij} = \frac{n_{i+}n_{+j}}{n}$

3.4 Mantel-Haenszel Chi-square statistic

Another form of the Chi-square statistic is the Mantel-Haenszel chi-square statistic, Q . For 2×2 contingency tables,

$$Q = \frac{(n_{11} - m_{11})^2}{v_{11}}$$

$$m_{11} = n_{1+}n_{+1}/n; v_{11} = \frac{n_{1+}n_{2+}n_{+1}n_{+2}}{n^2(n-1)}$$

i Note

Q , also known as the randomization chi-square, is based on the condition that all marginal totals are fixed and the probability distribution of random allocation of units to groups is given by the hypergeometric distribution. m_{11} and v_{11} are the mean and variance of the hypergeometric distribution based on the 2×2 table, respectively.

3.5 Pearson Chi-square vs. Mantel-Haenszel Chi-square

The relationship can also be approximated by the following relation:

$$Q_P = \frac{n}{n-1}Q$$

For large n , both the MH chi-square and the Pearson chi-square approach a chi-square distribution with 1 degree of freedom (χ_1^2).

 Warning

Rule of thumb for Q_P : The expected value for each cell should be at least 5, i.e. $m_{ij} \geq 5$ for all i, j

 Tip

Because the MH chi-square statistic comes from a hypergeometric distribution, the MH chi-square statistic can be used for small samples using exact calculations of p-values from the hypergeometric distribution. This is not possible with the Pearson chi-square. The MH chi-square statistic can also be extended for use in the presence of covariates/stratifying variables.

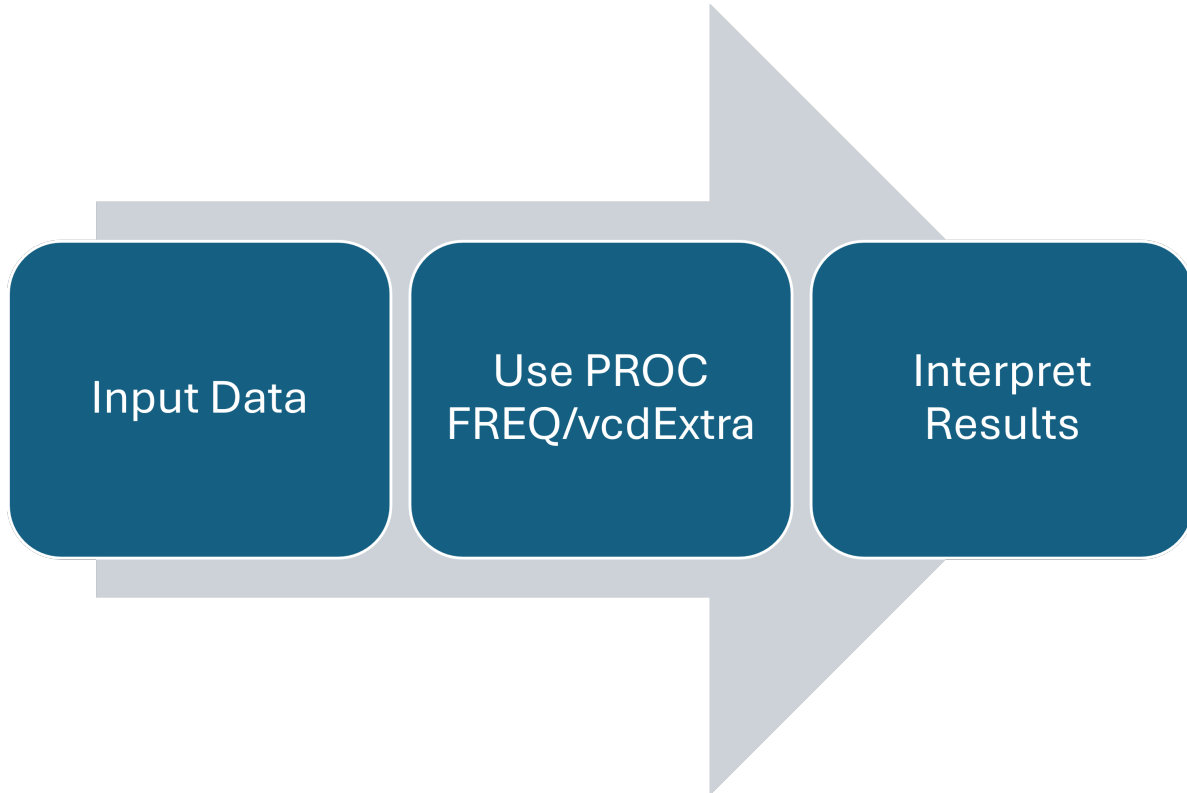
3.6 Hypothesis testing

The null hypothesis of no association states that there is no association between row and column variables. - Assuming this is true and assumptions are satisfied, Q and Q_P follow a χ^2_1 distribution for 2×2 tables.

 Note

The p-values are given by $P(\chi^2 > Q_P)$ and $P(\chi^2 > Q)$ for Pearson and MH Chi-squares, respectively.

3.7 Hypothesis Testing: SAS



3.8 SAS: FREQ procedure

- The **weight** statement tells SAS if there is an aggregated count variable present in the data set.
- The **tables** statement tells SAS to create a contingency table using the stated variables.
 - Under tables, you can include the **cmh** or the **chisq** option to calculate the randomization and Pearson chi-square values.

Shown below is a sample SAS code.

```
proc freq;  
weight count;  
tables a*b/ cmh chisq;  
run;
```

3.9 R: vcdExtra

In R, the `vcdExtra` package is required to perform the Mantel-Haenszel chi-square tests. The function `CMHtest()` provides the value of the Mantel-Haenszel statistic, while `chisq.test()` provides the Pearson chi-square statistic.

3.10 Example

The table shows the information from a randomized clinical trial that compared an experimental treatment to a placebo treatment for a respiratory disorder.

```
library(tidyverse)
dat <- expand_grid(Treatment=factor(c("Placebo","Test"),levels=c("Placebo","Test")),
                  Outcome = factor(c("Favorable","Unfavorable"),levels=c("Favorable","Unfavorable")))

dat$count = c(16,40,48,20)

xtabs(count ~ .,data=dat)
```

Treatment	Outcome	
	Favorable	Unfavorable
Placebo	16	48
Test	40	20

3.10.1 Questions

- What is the Pearson chi-square value?
- What is the Mantel-Haenszel chi-square value?
- Is there strong evidence of an association between treatment and outcome?

3.10.2 SAS Code

```
1 SAScode <- "  
2 data respire;  
3 input treat $ outcome $ count;  
4 datalines;  
5 test f 40  
6 test u 20  
7 placebo f 16  
8 placebo u 48
```

```

9 ;
10
11 proc freq data=respire order=data;
12 weight count;
13 tables treat*outcome/chisq;
14 run;
15 "

```

3.10.3 R Code

```

dat <- expand.grid(Treatment=c("Placebo","Test"),
                  Outcome = c("Favorable","Unfavorable")) |>
mutate(count = c(16,40,48,20))

xtabs(count ~ .,data=dat) -> dat_cc

```

Pearson Chi-square (without continuity adjustment)

```

#install.packages("tidyverse")
library(tidyverse)
dat <- expand.grid(Treatment=c("Placebo","Test"),
                  Outcome = c("Favorable","Unfavorable")) |>
mutate(count = c(16,40,48,20))

xtabs(count ~ .,data=dat)

```

	Outcome	
Treatment	Favorable	Unfavorable
Placebo	16	48
Test	40	20

```
chisq.test(dat_cc,correct=F)
```

Pearson's Chi-squared test

```

data: dat_cc
X-squared = 21.709, df = 1, p-value = 3.174e-06

```

Randomization Chi-square

```
# install.packages("vcdExtra")
library(vcdExtra)
CMH_CC <-CMHtest(dat_cc)
CMH_CC
```

Cochran-Mantel-Haenszel Statistics for Treatment by Outcome

	AltHypothesis	Chisq	Df	Prob
cor	Nonzero correlation	21.534	1	3.4768e-06
rmeans	Row mean scores differ	21.534	1	3.4768e-06
cmeans	Col mean scores differ	21.534	1	3.4768e-06
general	General association	21.534	1	3.4768e-06

3.10.4 Answers

- What is the Pearson chi-square value? **21.7**
- What is the Mantel-Haenszel chi-square value? **21.53**
- Is there strong evidence of an association between treatment and outcome?
 - $p < 0.0001$; There is strong evidence of an association between the test treatment and the outcome.

3.11 Exercise

Suppose we wish to determine if the frequency of falls is related to wheelchair use for Polio survivors. The data is shown in the table.

Fall	Wheelchair	
	Yes	No
Fallers	62	121
Nonfallers	18	32

3.11.1 Questions

- Write the code to enter the data and analyze using the FREQ procedure.
- What is the Pearson chi-square value?
- What is the Mantel-Haenszel chi-square value?
- Using a significance level of 0.05, do we have sufficient evidence to warrant the conclusion that wheelchair use and falling are associated?

3.11.2 SAS code

```
1 data wheelchair;
2 input fall $ wheelchair $ count;
3 datalines;
4 f yes 62
5 n yes 18
6 f no 121
7 n no 32
8 ;
9
10 proc freq data=wheelchair order=data;
11 weight count;
12 tables fall*wheelchair/chisq;
13 run;
```

3.11.3 R

Pearson Chi-square (without continuity adjustment)

```
#install.packages("tidyverse")

chisq.test(dat_cc, correct = F)
```

Pearson's Chi-squared test

```
data: dat_cc
X-squared = 0.078299, df = 1, p-value = 0.7796
```

Randomization Chi-square

```
# install.packages("vcdExtra")
CMH_CC <-CMHtest(dat_cc)
CMH_CC$table
```

	Chisq	Df	Prob
cor	0.07796307	1	0.780077
rmeans	0.07796307	1	0.780077
cmeans	0.07796307	1	0.780077
general	0.07796307	1	0.780077

4 Exact Methods

4.1 Fisher's Exact Test

The chi-square distribution assumption for Q and Q_P includes a large sample assumption.

Warning

What if $m_{ij} < 5$?

The **Fisher's Exact Test** can be used to analyze these contingency tables.

4.2 Fisher's Exact Test

Fisher's Exact Test is based on the hypergeometric distribution. The test calculates an *exact* p-value, unlike the chi-square test which only uses an approximation.

Note

The test is easy to calculate for 2×2 and total sample sizes below 50, but computationally expensive as the number increases.

The p-value can be calculated using the hypergeometric distribution with the following parameters: $Hyper(N = n, m = n_{+1}, k = n_{1+})$

4.3 Fisher's Exact Test: Hypotheses

Null hypothesis H_0 : there is no association between the row and column variables

- One sided H_a : There is a positive/negative association between the row and column variables. OR One combination of the row and column variable levels is more favored compared to the others.
- Two-sided H_a : there is an association between the row and column variables.

Note

To solve the exact p-value using the hypergeometric distribution, only add probabilities of outcomes that have lower probabilities than the observed value in the direction of the alternative hypotheses.

4.4 Fisher's Exact Test: SAS

Including the `exact` or the `chisq` option under the `tables` statement will yield the results of the exact test.

```
proc freq;
weight count;
tables a*b/ exact chisq;
run;
```

4.5 Fisher's Exact Test: R

The `fisher.test()` can perform Fisher's exact test in R using a contingency table as input.

```
fisher.test(contingency_table,alternative="two.sided")
```

4.6 Example

The table shown shows data from a study on treatments for healing severe infections. Calculate the exact p-value of this contingency table using the Fisher's Exact Test.

- Under the alternative hypothesis: The test treatment is positively associated with the outcome.
- Under the alternative hypothesis: The test treatment is associated with the outcome.
- At a significance level of 0.10, do you have enough evidence to conclude that the treatment is associated with the outcome?

Treatment	Outcome	
	Favorable	Unfavorable
Test	5	1
Control	1	3

4.7 Example

4.7.1 SAS Code

```

data severe;
input treat $ outcome $ count;
datalines;
Test f 5
Test u 1
Control f 1
Control u 3
;
proc freq order=data data=severe;
weight count;
tables treat*outcome / chisq nocol;
run;

```

4.7.2 R Code

```

dat <- expand.grid(Treatment=factor(c("Test","Control"),levels=c("Test","Control")),
                  Outcome = factor(c("Favorable","Unfavorable"),levels=c("Favorable","Unfavorable")),
                  dat$count <- c(5,1,1,3)

dat_cc <- xtabs(count ~ Treatment+Outcome,data=dat)
dat_cc

```

	Outcome	
Treatment	Favorable	Unfavorable
Test	5	1
Control	1	3

```

fisher.test(dat_cc, alternative = "greater") # one-sided

```

Fisher's Exact Test for Count Data

```

data: dat_cc
p-value = 0.119
alternative hypothesis: true odds ratio is greater than 1
95 percent confidence interval:
 0.5819      Inf
sample estimates:
odds ratio
 10.29391

```

```
fisher.test(dat_cc,alternative= "two.sided")
```

Fisher's Exact Test for Count Data

```
data:  dat_cc
p-value = 0.1905
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 0.40161 930.24692
sample estimates:
odds ratio
 10.29391
```

4.7.3 Manual p-values: 1-sided

- One-sided hypothesis (+) corresponds to the the probabilities $P(X \geq 5)$ which are equal to or less than $P(X = 5)$.

```
N <- 5+1+1+3
m <- 6
k <- 6

dhyper(x=5,m=m,n=N-m,k=k)
```

```
[1] 0.1142857
```

For $X > 5$, the only possible value of X is 6.

```
dhyper(x=6,m=m,n=N-m,k=k)
```

```
[1] 0.004761905
```

The p-value is then the sum of these two probabilities:

```
dhyper(x=5,m=m,n=N-m,k=k) + dhyper(x=6,m=m,n=N-m,k=k)
```

```
[1] 0.1190476
```

4.7.4 Manual p-values: 2-sided

- Two-sided hypothesis need to include values $P(X < 5)$ that have probabilities lower than $P(X = 5)$. These probabilities will be added to the one-sided p-value to get the two-sided p-value.

```
dhyper(x=4,m=m,n=N-m,k=k)
```

```
[1] 0.4285714
```

```
dhyper(x=3,m=m,n=N-m,k=k)
```

```
[1] 0.3809524
```

```
dhyper(x=2,m=m,n=N-m,k=k)
```

```
[1] 0.07142857
```

```
dhyper(x=1,m=m,n=N-m,k=k)
```

```
[1] 0
```

```
dhyper(x=0,m=m,n=N-m,k=k)
```

```
[1] 0
```

$P(X = 0)$, $P(X = 1)$, and $P(X = 2)$ have lower probabilities compared to $P(X = 5)$. Hence, we add these probabilities to the one-sided p-value to get the two-sided p-value.

```
dhyper(x=5,m=m,n=N-m,k=k) + dhyper(x=6,m=m,n=N-m,k=k) + dhyper(x=2,m=m,n=N-m,k=k) + dhyper(x=
```

```
[1] 0.1904762
```

4.7.5 Answers

- Under the alternative hypothesis: The test treatment is positively associated with the outcome. $p=0.119$
- Under the alternative hypothesis: The test treatment is associated with the outcome. $p=0.1905$
- At a significance level of 0.10, do you have enough evidence to conclude that the treatment is associated with the outcome? **At $\alpha = 0.10$, we do not have sufficient evidence to conclude that treatment is associated with the outcomes in healing severe infections.**

Note

In SAS, be careful when selecting which one-sided alternative hypothesis to interpret from the results!

4.8 Exercise

Fisher's experiment: A number of individuals gathered together for afternoon tea. A lady in the group said that she could differentiate between whether the milk or the tea was poured first into a cup. Fisher suggested to pour tea first into four cups and to pour milk first into four cups. The lady was blinded to which cups had milk or tea poured first, but she knew there were four of each. The contingency table is shown below. At $\alpha = 0.05$, do we have sufficient evidence to conclude that the lady can differentiate tea based on how they were poured?

4.8.1 Contingency Table

Guess		
Actual	Milk	Tea
Milk	3	1
Tea	1	3

4.8.2 SAS Code

```
data tea;
input actual$ guess $ count;
datalines;
milk milk 3
milk tea 1
tea milk 1
```

```
tea tea 3
;

proc freq data=tea order=data;
weight count;
tables actual*guess/chisq;
run;
```

Direction is specified, hence we need to use the “>” one-sided p-value.

4.8.3 R `fisher.exact()`

Direction is specified, hence we need to use the “>” one-sided p-value.

```
dat <- expand.grid(Actual=factor(c("Milk","Tea"),levels=c("Milk","Tea")),
                  Guess = factor(c("Milk","Tea"),levels=c("Milk","Tea")))

dat$count <- c(3,1,1,3)

dat_cc <- xtabs(count ~ .,data=dat)
dat_cc
```

	Guess	
Actual	Milk	Tea
Milk	3	1
Tea	1	3

```
fisher.test(dat_cc, alternative="greater")
```

Fisher's Exact Test for Count Data

```
data: dat_cc
p-value = 0.2429
alternative hypothesis: true odds ratio is greater than 1
95 percent confidence interval:
 0.3135693      Inf
sample estimates:
odds ratio
 6.408309
```

4.8.4 Hypergeometric: R

Direction is specified, hence we need to use the “>” one-sided p-value.

```
N <- 8
m <- 4
k <- 4

dhyper(x=3,m=m,n=N-m,k=k)
```

```
[1] 0.2285714
```

The only other possible value of X is $X = 4$.

```
dhyper(x=4,m=m,n=N-m,k=k)
```

```
[1] 0.01428571
```

To get the one-sided p-value, we need to add these two probabilities:

```
dhyper(x=3,m=m,n=N-m,k=k) + dhyper(x=4,m=m,n=N-m,k=k)
```

```
[1] 0.2428571
```

Note

Note that we needed to add $P(X = 4)$ because its value is less than that of $P(X = 3)$.

4.8.5 Answer

The resulting p-value is 0.2429. At significance level $\alpha = 0.05$, we fail to reject the null hypothesis. We have no sufficient evidence to conclude that the lady can differentiate tea based on how they were poured.

5 Difference in Proportions

5.1 Difference in Proportions

We are often interested in comparing two independent populations. For binary responses, we can compare proportions of success.

i Note

Recall from Intro to Stats course: we can estimate the confidence intervals and standardized statistics for the difference of two proportions.

5.2 Difference in Proportions: Estimation

Estimating the difference of the two proportions involves calculating the confidence intervals. There are different methods to calculate the confidence interval for the difference in proportion.

Table 2.5 2×2 Contingency Table

	Yes	No	Total	Proportion Yes
Group 1	n_{11}	n_{12}	n_{1+}	$p_1 = n_{11}/n_{1+}$
Group 2	n_{21}	n_{22}	n_{2+}	$p_2 = n_{21}/n_{2+}$
Total	n_{+1}	n_{+2}	n	

i Note

Independence of the two populations (Groups 1 and 2) mean that we have two independent binomial distributions.

5.3 Confidence Intervals: Wald

The commonly used confidence interval in estimating difference in proportions is the **Wald** confidence interval.

$$p_1 - p_2 \pm z_{1-\alpha/2} \sqrt{\frac{p_1(1-p_1)}{n_{1+}} + \frac{p_2(1-p_2)}{n_{2+}}}$$

⚠ Warning

This CI fails when p_1 or p_2 are close to the extreme values (0 and 1). this CI does not account for the discreteness of the data.

5.4 Confidence Intervals: Corrected Wald

We introduce a correction term to the Wald confidence interval to account for the continuity of the data.

$$p_1 - p_2 \pm z_{1-\alpha/2} \sqrt{\frac{p_1(1-p_1)}{n_{1+}} + \frac{p_2(1-p_2)}{n_{2+}} + 1/2(1/n_{1+} + 1/n_{2+})}$$

Warning

This CI still fails when p_1 or p_2 are close to the extreme values (0 and 1).

5.5 Confidence Intervals: Agresti-Caffo

We introduce a correction term to the Wald confidence interval to account for the extreme values.

$$p_1 - p_2 \pm z_{1-\alpha/2} \sqrt{\frac{\tilde{p}_1(1-\tilde{p}_1)}{n_{1+} + 2} + \frac{\tilde{p}_2(1-\tilde{p}_2)}{n_{2+} + 2}}$$

where

$$\tilde{p}_1 = (n_{11} + 1)/(n_{1+} + 2); \tilde{p}_2 = (n_{21} + 1)/(n_{2+} + 2)$$

Note

This correction keeps the coverage of the confidence interval at 95% for all possible values of p_1 and p_2 .

5.6 Other Confidence Intervals

Confidence Intervals	SAS code	R code	Adjustment Type
Wald	<code>wald</code>	“wald”	Default
Corrected Wald	<code>correct</code>	“waldcc”	Continuity correction
Agresti-Caffo	<code>ac</code>	“ac”	Improved Coverage
Miettinen-Nurminen	<code>mn</code>	“mn”	Uses goodness-of-fit to improve coverage
Newcombe	<code>newcombe</code>	“scorecc”	hybrid interval that does well for small counts

Confidence Intervals	SAS code	R code	Adjustment Type
Hauck-Anderson	ha	“ha”	Accounts for row totals

5.7 Confidence Intervals: SAS

- The FREQ procedure has all mentioned confidence intervals
 - The options included in the `tables` statement are included in the previous slide.
 - The `riskdiff` option of the `tables` statement allows us to specify which confidence intervals we need for the difference as well as the estimates for the group proportions.
 - The `alpha` option can be used to change the confidence level.

Sample SAS code:

```
1 proc freq;
2 weight count;
3 tables treat*outcome/riskdiff(correct cl=(mn ac)) alpha=0.05;
4 run;
```

5.8 Confidence Intervals: R

The `BinomDiffCI()` function in the `DescTools` package in R can calculate these confidence intervals.

```
# install.packages("DescTools")
library(DescTools)
BinomDiffCI(x1=n11, n1 = row1sum, x2 = n21, n2=row2sum,
             conf.level=0.95,
             method = c("ac", "wald", "waldcc", "score", "scorecc", "mn",
                        "mee", "blj", "ha", "hal", "jp")
             )
```

5.9 Example

The contingency table shows the information from a randomized clinical trial that compared an experimental treatment to a placebo treatment for a respiratory disorder.

	Outcome	
Treatment	Favorable	Unfavorable
Placebo	16	48
Test	40	20

5.9.1 Question

Calculate the 95% confidence interval for the difference in proportion of favorable outcomes for both treatment groups.

- Wald
- Corrected Wald
- Agresti-Caffo
- Miettinen-Nurminen

5.9.2 SAS Code

```
1 data respire;
2 input treat $ outcome $ count;
3 datalines;
4 test f 40
5 test u 20
6 placebo f 16
7 placebo u 48
8 ;
9
10 proc freq data=respire order=data;
11 weight count;
12 tables treat*outcome/riskdiff(cl=(mn ac)) alpha=0.05;
13 tables treat*outcome/riskdiff(correct) alpha=0.05;
14 run;
```

5.9.3 R Code

```
dat <- expand.grid(Treatment=c("Placebo","Test"),
                  Outcome = c("Favorable","Unfavorable"))
dat$count <- c(16,40,48,20)

dat_cc <- xtabs(count ~ .,data=dat)
dat_cc
```

	Outcome	
Treatment	Favorable	Unfavorable
Placebo	16	48
Test	40	20

```
library(DescTools)
```

Warning: package 'DescTools' was built under R version 4.6.1

```
BinomDiffCI(x1=40,  
            n1 = 40+20,  
            x2=16,  
            n2=16+48,  
            method=c('wald','waldcc','ac','mn'),  
            conf.level = 0.95)
```

	est	lwr.ci	upr.ci
wald	0.4166667	0.2570362	0.5762972
waldcc	0.4166667	0.2408903	0.5924430
ac	0.4166667	0.2455745	0.5618546
mn	0.4166667	0.2460145	0.5626554

5.9.4 Answers

The estimated difference in proportion is 0.4167 with the following 95% confidence intervals: - Wald: **0.2570-0.5763** - Corrected Wald: **0.2409-0.5924** - Agresti-Caffo: **0.2456-0.5416** - Miettinen-Nurminen **0.2460-0.5627**

5.10 Exercise

One of the most famous and largest clinical trials ever performed was in 1954. The data obtained from this RCT is shown in the contingency table.

Salk vaccine clinical trial results; data source is Francis et al. (1955, p. 25).

	Polio	Polio free	Total
Vaccine	57	200,688	200,745
Placebo	142	201,087	201,229
Total	199	401,775	401,974

5.10.1 Problem

Calculate the 90% confidence interval for the difference in proportion of polio-free children between the vaccine and placebo group.

- Corrected Wald
- Miettinen-Nurminen
- Newcombe

5.10.2 SAS Code

```
1 data vaxx;
2 length treat $7.;
3 input treat $ outcome $ count;
4 datalines;
5 vax n 200688
6 vax p 57
7 placebo n 201087
8 placebo p 142
9 ;
10
11 proc freq data=vaxx order=data;
12 weight count;
13 tables treat*outcome/riskdiff(correct cl=(mn newcombe))
14 alpha=0.10;
15 run;
```

5.10.3 R Code

```
library(DescTools)
BinomDiffCI(x1=142,
            n1=201087+142,
            x2=57,
            n2 = 57+200688,
            method=c('waldcc','mn','newcombe'),
            conf.level = 0.9)
```

	est	lwr.ci	upr.ci
waldcc	0.0004217214	0.0003013909	0.0005420518
mn	0.0004217214	0.0003084844	0.0005405825

5.10.4 Answer

The estimated difference in proportion is 0.0004.

- Corrected Wald (0.0003,0.0005)
- Miettinen-Nurminen (0.0003,0.0005)
- Newcombe (0.0003,0.0005)

6 Odds Ratio and Relative Risk

6.1 Odds Ratio (OR)

Odds of success

The ratio of probability of success and failure of an outcome is referred to as the odds.

$$Odds = p/(1 - p)$$

Odds Ratio

The ratio of the odds of success for Group 1 and Group 2 is referred to as the odds ratio.

$$OR = \frac{p_1/(1 - p_1)}{p_2/(1 - p_2)} = (n_{11}n_{22}/n_{21}n_{12})$$

The odds ratio is commonly used as an effect size for retrospective studies. This statistic is also used in logistic regression techniques, which will be discussed later.

6.2 Relative Risk (RR)

The ratio of the risk developing a particular condition for individuals with risk factors present and absent is called the *relative risk*.

$$RR = p_1/p_2 = \frac{(n_{11}/n_{1+})}{(n_{21}/n_{2+})}$$

The relative risk is commonly used as an effect size in prospective studies.

Note

For cross-sectional data, the relative risk is known as the prevalence ratio because the disease and risk factor are assessed at the same time.

6.3 Odds Ratio & Relative Risk

- When OR, $RR > 1$, Group 1 is **more likely** than Group 2 to record success.
- When OR, $RR < 1$, Group 1 is **less likely** to record success.
- When OR, $RR = 1$, there is **no association** between group and success because the two groups are equally likely to record success.

Note

When the condition is rare, the OR is approximately equal to the RR.

6.4 Odds Ratio & Relative Risk: SAS

- OR and RR are calculated using the measures option in the TABLES statement in PROC FREQ.
- For small sample sizes, the exact confidence limits can be calculated for the odds ratios.

```
1 proc freq;  
2 weight count;  
3 tables a*b/measures;  
4 exact or rr;  
5 run;
```

6.5 Odds Ratio & Relative Risk: R

The odds ratio and relative risk can be calculated using functions from the `epitools` package.

```
library(epitools)  
epitools::oddsratio(contingency_table,  
  conf.level = 0.95,  
  method=c("fisher","wald"),  
  rev=c("rows","columns","both")  
)
```

```
epitools::riskratio(contingency_table,
  conf.level = 0.95,
  method=c("fisher","wald"),
  rev=c("rows","columns","both")
)
```

💡 Tip

The `fisher` method provides exact confidence intervals for relative risk and odds ratio. The `fisher.test()` function provides an exact confidence interval for the odds ratio based on the hypergeometric distribution, but it does not include the relative risk.

💡 Tip

The `rev` argument allows us to re-arrange the contingency table without defining factors to establish row and column R. The order of the columns for `oddsratio` and `riskratio` should follow this order:

- Rows (top to bottom): No exposure (risk absent), Exposure (risk present)
- Cols (left to right): Control (Outcome=0), Case (Outcome=1)

6.6 Example

The data set `severe_long.csv` includes data from a study on treatments for severe infections.

6.6.1 Problem

Calculate the asymptotic and exact confidence limits for the odds ratio of favorable outcomes for test and control groups.

```
dat <- read.csv("severe_long.csv")
head(dat)
```

	Treatment	Outcome
1	Test	Favorable
2	Test	Favorable
3	Test	Favorable
4	Test	Favorable
5	Test	Favorable
6	Test	Favorable

6.6.2 SAS Code

Import `severe_long` as `work.severe_long` in SAS.

```
proc freq order=data data=work.severe_long;
tables treat*outcome;
exact or;
run;
```

Method	Value and CI
Asymptotic	10 (1.0256, 97.5005)
Exact	10 (0.6896, 166.3562)

The two methods lead to different conclusions.

6.6.3 R Code

Create the contingency table.

```
dat_cc <- xtabs(~Treatment+Outcome, data=dat)
dat_cc
```

	Outcome	
Treatment	Favorable	Unfavorable
Control	2	4
Test	10	2

Calculate the odds ratio and the confidence intervals.

```
library(epitools)
```

Warning: package 'epitools' was built under R version 4.6.1

Attaching package: 'epitools'

The following object is masked from 'package:vcdExtra':

`expand.table`

The following object is masked from 'package:vcd':

oddsratio

```
epitools::oddsratio(dat_cc,  
  method=c("wald"),  
  conf.level=0.95,  
  rev="columns" #Reverses the columns  
)
```

Warning in chisq.test(xx, correct = correction): Chi-squared approximation may be incorrect

\$data

	Outcome		
Treatment	Unfavorable	Favorable	Total
Control	4	2	6
Test	2	10	12
Total	6	12	18

\$measure

	odds ratio with 95% C.I.		
Treatment	estimate	lower	upper
Control	1	NA	NA
Test	10	1.025636	97.50049

\$p.value

	two-sided		
Treatment	midp.exact	fisher.exact	chi.square
Control	NA	NA	NA
Test	0.06119371	0.1070351	0.03389485

\$correction

[1] FALSE

attr("method")

[1] "Unconditional MLE & normal approximation (Wald) CI"

```
epitools::oddsratio(dat_cc,  
  method=c("fisher"),  
  conf.level=0.95,
```

```
rev="columns" #Reverses the columns,
)
```

Warning in `chisq.test(xx, correct = correction)`: Chi-squared approximation may be incorrect

\$data

	Outcome		
Treatment	Unfavorable	Favorable	Total
Control	4	2	6
Test	2	10	12
Total	6	12	18

\$measure

	odds ratio with 95% C.I.		
Treatment	estimate	lower	upper
Control	1.000000	NA	NA
Test	8.457238	0.6896384	166.4345

\$p.value

	two-sided		
Treatment	midp.exact	fisher.exact	chi.square
Control	NA	NA	NA
Test	0.06119371	0.1070351	0.03389485

\$correction

[1] FALSE

attr("method")

[1] "Conditional MLE & exact CI from 'fisher.test'"

i Note

We report the **estimate** of the odds ratio from the "wald" method, but use the "fisher" confidence interval as the exact confidence interval.

We also use `epitools::oddsratio` to tell R that we are using the `oddsratio` function in `epitools` and not another package (`vcd` also has a package of the same name).

6.7 Exercise

Two groups having the same symptoms of respiratory illness were sampled. One group was treated with an experimental treatment and the other group was treated with a placebo. The participants were asked whether there was an overall treatment in overall respiration.

6.7.1 Data

	Outcome	
Treatment	Yes	No
Test	29	16
Placebo	14	31

6.7.2 SAS Code

```
data resp;
input Treatment $ Outcome $ count;
datalines;
Test Yes 29
Placebo Yes 14
Test No 16
Placebo No 31
;
proc freq data=resp order=data;
weight count;
tables treatment*outcome/measures;
run;
```

6.7.3 R Code

```
dat <- expand.grid(Treatment=c("Test","Placebo"),
                  Outcome = c("Yes","No"))
dat$count = c(29,14,16,31)

dat_cc<- xtabs(count ~ .,data=dat)
dat_cc
```

	Outcome	
Treatment	Yes	No

Test	29	16
Placebo	14	31

Odds ratio

```
library(epitools)
or <- epitools::oddsratio(dat_cc,
                           method="wald",
                           conf.level=0.95,
                           rev="both")
or
```

\$data

	Outcome		
Treatment	No	Yes	Total
Placebo	31	14	45
Test	16	29	45
Total	47	43	90

\$measure

	odds ratio with 95% C.I.		
Treatment	estimate	lower	upper
Placebo	1.000000	NA	NA
Test	4.013393	1.668039	9.656441

\$p.value

	two-sided		
Treatment	midp.exact	fisher.exact	chi.square
Placebo	NA	NA	NA
Test	0.001783016	0.002917802	0.001548669

\$correction

[1] FALSE

attr(,"method")

[1] "Unconditional MLE & normal approximation (Wald) CI"

or\$measure

	odds ratio with 95% C.I.		
Treatment	estimate	lower	upper

Placebo	1.000000	NA	NA
Test	4.013393	1.668039	9.656441

Relative Risk

```
rr <- riskratio(dat_cc,
                method="wald",
                rev="both")
rr
```

\$data

	Outcome		
Treatment	No	Yes	Total
Placebo	31	14	45
Test	16	29	45
Total	47	43	90

\$measure

	risk ratio with 95% C.I.		
Treatment	estimate	lower	upper
Placebo	1.000000	NA	NA
Test	2.071429	1.274194	3.367474

\$p.value

	two-sided		
Treatment	midp.exact	fisher.exact	chi.square
Placebo	NA	NA	NA
Test	0.001783016	0.002917802	0.001548669

\$correction

[1] FALSE

attr("method")

[1] "Unconditional MLE & normal approximation (Wald) CI"

rr\$measure

	risk ratio with 95% C.I.		
Treatment	estimate	lower	upper
Placebo	1.000000	NA	NA
Test	2.071429	1.274194	3.367474

6.7.4 Answer

- Calculate the odds ratio of respiratory improvement between test and placebo groups. **4.01 (1.67-9.66)**
- What is the relative risk of having respiratory improvement if we consider the treatment as a risk factor? **2.07 (1.27-3.37)**

7 Sensitivity and specificity

7.1 Sensitivity

True proportion of positive results that a test elicits when performed on subjects known to have the disease.

$$Sens = TP / (TP + FN) = n_{11} / n_{1+}$$

Sensitivity is the conditional probability of a positive test given that the subject has the disease ($P(PT|HD)$).

7.2 Specificity

True proportion of negative results that a test elicits when performed on subjects known to be disease free.

$$Sens = TN / (TN + FP) = n_{22} / n_{2+}$$

Specificity is the conditional probability of a negative test when the subject does not have the disease ($P(NT|ND)$).

7.3 Sensitivity and Specificity: SAS

The FREQ procedure can also calculate sensitivity and specificity.

- `riskdiff` approximates these metrics using the risk estimates.
 - Sensitivity: Row 1 risk estimate for Column 1
 - Specificity: Row 2 risk estimate for Column 2

7.4 Sensitivity and Specificity: R

The `epiR` package has the `epi.tests()` function that calculates sensitivity and specificity. For the following contingency table:

	Outcome +	Outcome -
Test +	n_{11}	n_{12}
Test -	n_{21}	n_{22}

```
install.packages("epiR")
library(epiR)

epi.tests(contingency_table,
          method=c("exact", "wilson", "agresti", "clopper-pearson", "jeffreys"))
```

Note

The default method used to calculate confidence intervals for the sensitivity and specificity is the `exact` method.

7.5 Example

The following table shows the results of a study investigating a new screening device for a skin disease. Calculate the sensitivity and specificity of the screening device with the corresponding exact 95% confidence interval.

7.5.1 Data

Test	Status	
	Disease Present	Disease Absent
Positive	52	20
Negative	8	100

7.5.2 SAS Code

```

data screening;
input disease $ test $ count @@;
datalines;
present + 52 present - 8
absent + 20 absent - 100
;
proc freq data=screening order=data;
weight count;
tables disease*test / riskdiff;
run;

```

7.5.3 R Code

```

dat <- expand.grid(Status=c("Disease Present","Disease Absent"),
                  Test = c("Positive","Negative")) |>
mutate(count = c(52,20,8,100))

xtabs(count ~ Test+Status,data=dat) -> dat_cc
dat_cc

```

	Status	
Test	Disease Present	Disease Absent
Positive	52	20
Negative	8	100

```
library(epiR)
```

Warning: package 'epiR' was built under R version 4.6.1

Loading required package: survival

Attaching package: 'survival'

The following object is masked from 'package:epitools':

ratetable

Package epiR 2.0.96 is loaded

Type `help(epi.about)` for summary information

Type `browseVignettes(package = 'epiR')` to learn how to use epiR for applied epidemiological a

```
epi.tests(dat_cc)
```

	Outcome +	Outcome -	Total
Test +	52	20	72
Test -	8	100	108
Total	60	120	180

Point estimates and 95% CIs:

Apparent prevalence *	0.40 (0.33, 0.48)
True prevalence *	0.33 (0.26, 0.41)
Sensitivity *	0.87 (0.75, 0.94)
Specificity *	0.83 (0.75, 0.90)
Positive predictive value *	0.72 (0.60, 0.82)
Negative predictive value *	0.93 (0.86, 0.97)
Positive likelihood ratio	5.20 (3.44, 7.85)
Negative likelihood ratio	0.16 (0.08, 0.31)
False T+ proportion for true D- *	0.17 (0.10, 0.25)
False T- proportion for true D+ *	0.13 (0.06, 0.25)
False T+ proportion for T+ *	0.28 (0.18, 0.40)
False T- proportion for T- *	0.07 (0.03, 0.14)
Correctly classified proportion *	0.84 (0.78, 0.89)

* Exact CIs

7.5.4 Answer

- Sensitivity: 0.87 (0.75-0.94)
- Specificity: 0.83 (0.75-0.90)

7.6 Exercise

A pharmaceutical company developed a screening test that can be used to detect infection of the Lassa virus. The newly developed test was said to be cheaper than the tests currently available to the public, but its performance was yet to be evaluated. The screening test was applied to 115 diagnosed cases (w/ Lassa virus) and 120 control participants (w/o Lassa virus) of the Lassa fever. The resulting contingency table is shown in the table.

7.6.1 Question

- What is the sensitivity of the test?
- What is the specificity of the test?

Status	Test	
	Positive	Negative
Disease Present	97	18
Disease Absent	45	75

7.6.2 R Code

```
library(tidyverse)
dat <- expand_grid(Status=c("Disease Present","Disease Absent"),
                  Test = c("Positive","Negative"))
dat$count = c(97,45,18,75)

dat_cc <- xtabs(count ~ Test+Status,data=dat)
dat_cc
```

Test	Status	
	Disease Present	Disease Absent
Positive	97	45
Negative	18	75

```
library(epiR)
epi.tests(dat_cc)
```

	Outcome +	Outcome -	Total
Test +	97	45	142
Test -	18	75	93

Total	115	120	235
-------	-----	-----	-----

Point estimates and 95% CIs:

Apparent prevalence *	0.60 (0.54, 0.67)
True prevalence *	0.49 (0.42, 0.56)
Sensitivity *	0.84 (0.76, 0.90)
Specificity *	0.62 (0.53, 0.71)
Positive predictive value *	0.68 (0.60, 0.76)
Negative predictive value *	0.81 (0.71, 0.88)
Positive likelihood ratio	2.25 (1.76, 2.87)
Negative likelihood ratio	0.25 (0.16, 0.39)
False T+ proportion for true D- *	0.38 (0.29, 0.47)
False T- proportion for true D+ *	0.16 (0.10, 0.24)
False T+ proportion for T+ *	0.32 (0.24, 0.40)
False T- proportion for T- *	0.19 (0.12, 0.29)
Correctly classified proportion *	0.73 (0.67, 0.79)

* Exact CIs

7.6.3 SAS Code

```
data screening;
input disease $ outcome $ count @@;
datalines;
present + 97 present - 18
absent + 45 absent - 75
;
proc freq data=screening order=data;
weight count;
tables disease*outcome / riskdiff;
run;
```

7.6.4 Answer

- What is the sensitivity of the test? **0.8435 (0.76, 0.90)**
- What is the specificity of the test? **0.6250 (0.53, 0.71)**

8 McNemar's Test

8.1 McNemar's Test

When the 2x2 table provides information from matched pairs, the two responses are related to each other. Hence, independence cannot be assumed.

i Null Hypothesis

H_0 : The marginal probabilities of each outcome are the same.

i Test Statistic

The test statistic for the McNemar's test is Q_M given by

$$Q_M = (n_{12} - n_{21})^2 / (n_{12} + n_{21})$$

8.2 McNemar's Test: SAS

- The **agree** option in the **tables** statement can be used to perform the McNemar's test.
- Exact p-values can be calculated using the **exact** statement.

8.2.1 Sample SAS Code

```
proc freq;  
weight count;  
tables a*b/agree;  
exact mcnem;  
run;
```

8.3 McNemar's Test: R

The `mcnemar.test()` function can be used to perform the McNemar's test.

```
mcnemar.test(contingency_table) # no continuity correction
```

Note

The p-value calculated using this function adds the continuity correction by default.

8.4 Example

The table displays data collected by political science researchers who polled husbands and wives on whether they approved of one of their U.S. senators. The cell counts represent the number of pairs of husbands and wives who fit the configurations indicated by the row and column levels. Use the McNemar's test to determine whether there is no difference in husband and wife approvals on the state senator.

8.4.1 Data

		Wife	
Husband		Yes	No
Yes		20	5
No		10	10

8.4.2 SAS Code

```
1 data approval;
2 input hus_resp $ wif_resp $ count;
3 datalines;
4 yes yes 20
5 yes no 5
6 no yes 10
7 no no 10
8 ;
9
10 proc freq order=data;
11 weight count;
12 tables hus_resp*wif_resp / agree;
13 exact mcnem;
14 run;
```

8.4.3 R Code

```
dat <- expand.grid(Husband=c("Yes","No"),
                  Wife = c("Yes","No"))
dat$count <- c(20,10,5,10)

dat_cc <- xtabs(count ~ Husband+Wife,data=dat)
dat_cc
```

	Wife	
Husband	Yes	No
Yes	20	5
No	10	10

```
mcnemar.test(dat_cc)
```

McNemar's Chi-squared test with continuity correction

```
data: dat_cc
McNemar's chi-squared = 1.0667, df = 1, p-value = 0.3017
```

i Note

If we want to recover the asymptotic p-value from SAS, we need to remove the continuity correction.

```
mcnemar.test(dat_cc,correct=F)
```

McNemar's Chi-squared test

```
data: dat_cc
McNemar's chi-squared = 1.6667, df = 1, p-value = 0.1967
```

8.4.4 Answer

The exact/corrected p-value = 0.3018, which implies that there is no evidence of a difference between the senator approval ratings of the husbands and wives.

8.5 Exercise

The tables show the data from a study that compared the diagnostic accuracy of magnetic resonance imaging (MRI) and ultrasound in patients who had been established as having localized prostate cancer by a gold standard test.

8.5.1 Question

Use the McNemar's test to determine whether the two methods have a different probabilities in diagnosing localized prostate cancer. (*Hint: Small cell counts*)

MRI	Ultrasound	
	Localized	Advanced
Localized	4	6
Advanced	3	3

8.5.2 R Code

MRI	Ultrasound	
	Localized	Advanced
Localized	4	6
Advanced	3	3

```
mcnemar.test(dat_cc)
```

McNemar's Chi-squared test with continuity correction

```
data: dat_cc
```

```
McNemar's chi-squared = 0.44444, df = 1, p-value = 0.505
```

8.5.3 SAS Code

```
data imaging;
input mri $ ult $ count;
datalines;
loc loc 4
loc adv 6
adv loc 3
adv adv 3
```

```
;
proc freq order=data;
weight count;
tables mri*ult / agree;
exact mcnem;
run;
```

8.5.4 Answer

Use the McNemar's test to determine whether the two methods have a different probabilities in diagnosing localized prostate cancer. (*Hint: Small cell counts*)

- Use exact p-value
- $p = 0.51$. No evidence of any difference in probabilities in correctly diagnosing localized prostate cancer.

9 Incidence Densities

9.1 Incidence Densities

- Counts of subjects who responded within a duration of exposure to an event.
- Poisson distribution: counts
- Examples:
 - Bacteria colony count
 - Disease incidence
 - Accidents
 - Vaccine failure
- Common measure of duration of exposure/ time-at-risk: person-time

9.2 Poisson Distribution

Recall that the Poisson distribution is described by the following density function:

$$P(x) = e^{-\lambda} \lambda^x / x!$$

where λ is the expected number of events at a given unit of time.

- is also referred to as incidence density.
- For t units of time, the expected number of events scales to λt .
- Main assumption: The time to event is a random variable following an exponential distribution.

9.3 Incidence Density Ratios

- The table shows the number of infections (events) and person years for participants of an RCT.
 - n_v and n_p are Poisson random variables with respective incidence densities equal to λ_v and λ_p
- If λ_v is the incidence density for the vaccine group, then the expected number of events is $\lambda_v N_v$.
- Similarly, the expected number of events in the placebo group is $\lambda_p N_p$
- We are interested in testing the null hypothesis $\lambda_v/\lambda_p = 1$.
 - λ_v/λ_p is referred to as the **Incidence Density Ratio (IDR)**

Table 2.14 Vaccine Study

Treatment	Events	Person Years
Vaccine	n_v	N_v
Placebo	n_p	N_p

9.4 Incidence Density Ratios

The IDR can be estimated using conditional binomial distributions assuming n_v and n_p are independent Poisson processes.

$$P(n_v | n_v + n_p) \sim \text{Bin} \left(N = n_v + n_p, p = \frac{\lambda_v N_v}{\lambda_v N_v + \lambda_p N_p} \right)$$

9.5 Incidence Density Ratios: Confidence Interval

The binomial proportion can be estimated using the proportion of events in the treatment group:

$$p = \frac{n_v}{n_v + n_p} = \frac{IDR(N_v/N_p)}{1 + (IDR)(N_v/N_p)}$$

Using algebra (Problem Set 1), *it is easy to show that*:

$$IDR = \frac{p}{1-p} (N_p/N_v)$$

9.6 Incidence Density Ratios: Confidence Interval

You can calculate the confidence interval estimates for p using the binomial distribution assumption (p_L, p_U) and apply the scaling factor to each confidence limit.

$$95\%CI : \left(\frac{p_L}{1-p_L} (N_p/N_v), \frac{p_U}{1-p_U} (N_p/N_v) \right)$$

9.7 Incidence Density Ratios: SAS

- The `binomial` option in the `tables` statement produces the estimate and exact confidence limits for the binomial proportion p , i.e. p_L and p_U .
- We use the estimate and exact confidence limits of p in calculating the IDR.
- The `IML` procedure, which is the procedure used to perform mathematical equations, is used to calculate the IDR confidence limits.
 - Later on, we will use the `GENMOD` procedure to calculate exact confidence intervals for the IDR.

9.8 Incidence Density Ratios: R

The `epitools` package includes the `rateratio()` function that can calculate an approximation of the incidence density ratio.

```
rateratio(contingency_table)
#OR

rateratio(x=events,y=person_years)
```

9.9 Example

The table shows the results of an international study of the effect of a rotavirus vaccine in nearly 70,000 infants on health care events. The events counted were hospitalizations or emergency room visits.

9.9.1 Questions

- Estimate the IDR of hospitalization/emergency room visits between the two groups.
- Calculate the 95% CI.

Table 2.13 Vaccine Study Results for U.S.

Treatment	Events	Person Years
Vaccine	3	7500
Placebo	58	7250

9.9.2 SAS Code

```
1 data vaccine2;
2 input Outcome $ Count @@;
3 datalines;
4 vaccine 3 placebo 58
5 ;
6
7 ods select BinomialCLs; *this statement filters the output.;
8
9
10 proc freq order=data data=vaccine2;
11 weight count;
12 tables Outcome / binomial (exact); *calculates the exact confidence limits for the binomial p
13 ods output BinomialCLs=BinomialCLs; *outputs a dataset that includes the binomial CLs.;
14 run;
```

IDR	IDR_lower	IDR_higher
0.05	0.01002014	0.1535468

```

15
16 proc iml ;
17 Use BinomialCLs var{LowerCL UpperCL};
18 read all into CL;
19 print CL;
20 q = { 3, 7500, 58 , 7250 }; *vector of numbers from the table;
21 C= q[2]/q[4]; *q[i] represents the ith number in the q vector defined above. C is the person
22 P= q[1]/ (q[1] + q[3]); *estimate of the binomial distribution;
23 R= P/ ((1-P) *C) ; * IDR estimate;
24 CI= CL[1]/((1-CL[1])*C) || CL[2]/((1-CL[2])*C) ; *IDR CL;
25 print r CI;
26 quit;

```

① order=data is important so as to keep the proportions consistent.

9.9.3 R Code

```

events <- c(58,3)
person_years <- c(7250,7500)

mult <- person_years[1]/person_years[2] #multiplier
DescTools::BinomCI(events[2],sum(events),method = "clopper-pearson") -> p_est

data.frame(IDR = p_est[1]/(1-p_est[1])*mult,
           IDR_lower = p_est[2]/(1-p_est[2])*mult,
           IDR_higher = p_est[3]/(1-p_est[3])*mult) |> gt::gt()

```

```
rateratio(x=events,y=person_years)
```

```
$data
```

```

      Outcome
Predictor  events person_years
Exposed1    58      7250

```

Exposed2	3	7500	
Total	61	14750	


```

$measure
      rate ratio with 95% C.I.
Predictor  estimate      lower      upper
Exposed1  1.00000000      NA        NA
Exposed2  0.05253473  0.01239861  0.1418922

$p.value
      two-sided
Predictor      midp.exact      wald
Exposed1      NA        NA
Exposed2  0.000000000000006708884  0.00000000000007207568

attr(,"method")
[1] "Median unbiased estimate & mid-p exact CI"

```

9.9.4 Answers

- Estimate the IDR of hospitalization/emergency room visits between the two groups. **IDR = 0.05**
- Calculate the 95% CI. **(0.01-0.15)** or with `riskratio`: (0.01-0.14)

9.10 Exercise

The table lists the number of accidents for day and night shift workers at a certain factory in Southeast Asia. The company wants you to investigate if there is any evidence of night shift workers having a higher accident density, i.e., higher incidence density of accidents, compared to the day shift.

9.10.1 Data

Shift	Metric	
	Accidents	Person-Hours
Night	25	1150
Day	20	1240

9.10.2 SAS Code

```
data accident;
/*data only has the events. Person-Hours are used later.*/
input Outcome $ Count @@;
datalines;
night 25 day 20
;

ods select BinomialCLs; *this statement filters the output.;

proc freq order=data data=accident;
weight count;
tables Outcome / binomial (exact) alpha=0.05; *calculates the exact confidence limits for the
ods output BinomialCLs=BinomialCLs; *outputs a dataset that includes the binomial CLs.;
run;

proc iml ;
Use BinomialCLs var{LowerCL UpperCL};
read all into CL;
print CL;
q = { 25 1150 20 1240 }; *vector of numbers from the table;
C= q[2]/q[4]; *q[i] represents the ith number in the q vector defined above. C is the person
P= q[1]/ (q[1] + q[3]); *estimate of the binomial distribution;
R= P/ ((1-P) *C) ; * IDR estimate;
CI= CL[1]/((1-CL[1])*C) || CL[2]/((1-CL[2])*C) ; *IDR CL;
print r CI;
quit;
```

① “Success” is an event happening during the night shift.

9.10.3 R Code

```
events <- c(20,25)
person_years <- c(1240,1150)

mult <- person_years[1]/person_years[2] #multiplier
DescTools::BinomCI(events[2],sum(events),method = "clopper-pearson") -> p_est
```

IDR	IDR_lower	IDR_higher
1.347826	0.7187612	2.559057

```
data.frame(IDR = p_est[1]/(1-p_est[1])*mult,
            IDR_lower = p_est[2]/(1-p_est[2])*mult,
            IDR_higher = p_est[3]/(1-p_est[3])*mult) |> gt::gt()
```

Using rateratio()

```
rateratio(x=events,y=person_years)
```

\$data

	Outcome	
Predictor	events	person_years
Exposed1	20	1240
Exposed2	25	1150
Total	45	2390

\$measure

	rate ratio with 95% C.I.		
Predictor	estimate	lower	upper
Exposed1	1.000000	NA	NA
Exposed2	1.345627	0.7466559	2.45738

\$p.value

	two-sided	
Predictor	midp.exact	wald
Exposed1	NA	NA
Exposed2	0.3232959	0.3179524

attr(,"method")

[1] "Median unbiased estimate & mid-p exact CI"

9.10.4 Answer

- IDR: 1.35 (0.72-2.56)
- with riskratio: (0.75-2.46)

10 Power Analysis

10.1 Power Analysis

- Power is defined as the probability of rejecting the null hypothesis when it is false.
 - Acceptable range: 80% and above.
- Also known as (a priori) Sample Size Calculations
 - How many samples do I need to detect a set difference between the treatment and placebo group at a significance level of 0.05?

10.2 Power Analysis: SAS

- The `POWER` procedure enables us to calculate how many samples are needed to detect a set effect size in examining the difference between two proportions from two different populations.
- Specifically, the `TWOSAMPLEFREQ` option
 - RCT: treatment vs. control
- We need to set an effect size (theory-based), significance level, and power level.
 - Other options can be found [here](#)

10.3 Example

Suppose prior literature states that the incidence rate of a chronic disease is 0.40. Suppose we want to perform a prospective study to test the association between the disease incidence and family history, i.e. whether other family members were diagnosed with the chronic disease. We want to detect a proportion difference of 0.10 between groups with power of 80% (0.80) at a significance level of 0.05.

10.4 Example

Suppose prior literature states that the incidence rate of a chronic disease is 0.40. Suppose we want to perform a prospective study to test the association between the disease incidence and family history, i.e. whether other family members were diagnosed with the chronic disease. We want to detect a proportion difference of 0.10 between groups with power of 80% (0.80) at a significance level of 0.05.

- We need 388 units per group (family +, family -). This brings us to 776 total units.

10.4.1 SAS Code

```
proc power ;  
twosamplefreq  
    refproportion=0.4  
    alpha=0.05  
    power=0.80  
    proportiondiff=0.1  
    npergroup=.;  
  
run;
```

10.4.2 R Code

```
power.prop.test(p1=0.4,p2=0.5,power=0.8)
```

Two-sample comparison of proportions power calculation

```
      n = 387.3385  
    p1 = 0.4  
    p2 = 0.5  
sig.level = 0.05  
  power = 0.8  
alternative = two.sided
```

NOTE: n is number in *each* group